CCW® Armored Power, 5 kV 133%/8 kV 100%, Shielded, 3/C VFD

UL Type MC-HL or MV-105, EPR, 105°C, Cable Tray Use, Sunlight-Resistant
Direct Burial, ABS CWCMC



Product Construction:

Conductor:
- Bare annealed copper per ASTM B3
- Compact stranding per ASTM B496

Extruded Strand Shield (ESS):
- Extruded thermoset semi-conductor stress control layer over the conductor per ICEA S-93-639 and UL 1072

Insulation:
- 115 mils Ethylene Propylene Rubber (EPR) insulation per ICEA S-93-639 and UL 1072

Extruded Insulation Shield (EIS):
- Thermoset semi-conducting polymeric layer, free stripping from the insulation per ICEA S-93-639 and UL 1072

Shield:
- 5 mil annealed bare copper tape with 25% overlap

Phase Identification:
- Color-coded polymeric identification tape laid under the shield — black, red and blue

Grounding Conductors:
- Three (3) split Class B stranded bare annealed copper grounding conductors
- Sized in accordance with UL 1072 and NEC Article 250

Cable Assembly:
- Insulated and grounding conductors are cabled together with non-hygroscopic fillers when required
- Binder tape is applied over the cabled core

CCW Armor:
- Impervious, continuously welded and corrugated aluminum alloy sheath per UL 1072 and UL 1569
- CCW armor conductivity meets the grounding requirements of NEC Article 250

Jacket:
- Flame-retardant, moisture- and sunlight-resistant Polyvinyl Chloride (PVC), yellow
- Low temperature performance meets ASTM D746 brittleness temperature at or below -40°C

Applications:

- Variable Frequency Drives: 3-conductor CCW armored cables with three (3) symmetrical grounding wires are the preferred wiring method for use with AC motors controlled by pulse-width modulated inverters in VFD applications
- CCW armored Medium-Voltage power cables offer an economical, rugged and reliable alternative to labor-intensive cable in conduit wiring methods
- For use on feeders and branch circuits in industrial power distribution systems per NEC Articles 328 and 330
- For use in Class I, II and III, Divisions 1 and 2; and Class I, Zones 1 and 2 hazardous locations per NEC Articles 501, 502, 503 and 505
- Installed on metal racks, troughs, in raceways, in cable trays or secured to supports spaced not more than six feet apart
- Installed in both exposed and concealed work, wet or dry locations, directly buried or embedded in concrete
- Recognized for use on fixed or floating offshore petroleum facilities as recommended by the American Petroleum Institute

Features:

- CCW armor provides an impervious barrier to moisture, gas and liquids and meets the grounding requirements of UL 1072 and the NEC

Features: (cont'd)

- Triple Extrusion: The strand shield, EPR insulation and insulation shield are all extruded in one operation
- General Cable's EPR insulation system has outstanding corona resistance and high dielectric strength, and it provides electrical stability under stress
- Cable meets cold impact at -40°C
- 105°C continuous operating temperature, wet or dry
- 140°C emergency rating
- 250°C short circuit rating

Specifications:

Design Adherence:
- ICEA S-93-639/WC-74, 5-46 kV Shielded Power Cable
- AEIC CS8 Specification for Shielded Power Cable, 5-46 kV
- UL 1072 Medium-Voltage Power Cables
- UL 1569 Metal Clad Cables
- UL 2225 Cables and Cable Fittings for Use in Hazardous Locations
- UL 1309 Marine Shipboard Cable
- CSA C68.3

Flame Tests:
- ICEA T-29-520 (210,000 BTU/hr)
- IEEE 383 (70,000 BTU/hr)
- CSA FT4
- IEEE 1202 (70,000 BTU/hr)
- UL 1072
- IEC 60332-3 Category A

Compliances:
- UL Type MV-105, UL File # E90501
- UL Type MC-HL, SUN RES, CT USE, DIR BUR, -40°C, FT4, UL File # E90496
- UL Listed Marine Shipboard, UL File # E85994
- American Bureau of Shipping (ABS) Listed for CWCMC

General Cable
Phone: 888-593-3355
www.generalcable.com

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PART NUMBER	COND. SIZE AWG (kcmil)	NO. OF COND.	INSULATION THICKNESS mils	INSULATION THICKNESS mm	NOMINAL O.D. OVER INSULATION INCHES	NOMINAL O.D. OVER INSULATION mm	BARE GROUND AWG	NOMINAL CORE O.D. INCHES	NOMINAL CORE O.D. mm	NOMINAL ARMOR O.D. INCHES	NOMINAL ARMOR O.D. mm	JACKET THICKNESS mils	JACKET THICKNESS mm	NOMINAL OVERALL O.D. INCHES	NOMINAL OVERALL O.D. mm	APPROXIMATE NET WEIGHT LBS/ 1000 FT	APPROXIMATE NET WEIGHT kg/ 1000 m	5 kV[3] AMPACITY IN AIR[1]	5 kV[3] AMPACITY DIRECT BURIAL[2]
3/C WITH GROUND MC-HL OR MV-105, 115 MILS EPR, 5 kV 133% AND 8 kV 100% INSULATION LEVEL																			
9800.00603310	6 (7/W) (13.3 mm²)	3	115	2.9	0.43	10.9	3 x #10	1.15	29.2	1.37	34.8	50	1.27	1.48	37.6	1,121	1,668	88	115
9800.00403310	4 (7/W) (21.2 mm²)	3	115	2.9	0.48	12.2	3 x #10	1.24	31.5	1.51	38.4	60	1.52	1.65	41.9	1,418	2,110	115	150
9800.00203310	2 (7/W) (33.6 mm²)	3	115	2.9	0.53	13.5	3 x #10	1.37	34.8	1.64	41.7	60	1.52	1.78	45.2	1,731	2,576	154	190
9800.00103308	1 (19/W) (42.4 mm²)	3	115	2.9	0.57	14.5	3 x #8	1.47	37.3	1.69	42.9	60	1.52	1.82	46.2	1,978	2,944	180	215
9800.11003308	1/0 (19/W) (53.5 mm²)	3	115	2.9	0.60	15.2	3 x #8	1.56	39.6	1.78	45.2	60	1.52	1.91	48.5	2,259	3,362	205	245
9800.21003308	2/0 (19/W) (67.4 mm²)	3	115	2.9	0.64	16.3	3 x #8	1.61	40.9	1.92	48.8	60	1.52	2.05	52.1	2,626	3,908	240	280
9800.41003307	4/0 (19/W) (107 mm²)	3	115	2.9	0.74	18.8	3 x #7	1.82	46.2	2.15	54.6	60	1.52	2.28	57.9	3,650	5,432	320	360
9800.25003306	250 (37/W) (127 mm²)	3	115	2.9	0.80	20.3	3 x #6	2.01	51.1	2.23	56.6	60	1.52	2.36	59.9	4,060	6,042	355	395
9800.35003306	350 (37/W) (177 mm²)	3	115	2.9	0.89	22.6	3 x #6	2.10	53.3	2.45	62.2	75	1.91	2.61	66.3	5,045	7,508	440	475
9800.50003305	500 (37/W) (253 mm²)	3	115	2.9	1.01	25.7	3 x #5	2.39	60.7	2.75	69.9	75	1.91	2.92	74.2	7,137	10,621	545	570
9800.75003304	750 (61/W) (380 mm²)	3	115	2.9	1.19	30.2	3 x #4	3.07	78.0	3.32	84.3	85	2.16	3.50	88.9	10,268	15,280	685	700
9800.10003304	1000 (61/W) (507 mm²)	3	115	2.9	1.34	34.0	3 x #4	3.43	87.1	3.76	95.5	85	2.16	3.94	100.1	13,051	19,422	790	785

Dimensions and weights are nominal; subject to industry tolerances.
[1] In-air ampacities are per NEC Table 310.60(C)(71) for three insulated copper conductors rated 105°C, cabled with an overall covering and isolated in air at 40°C ambient temperature.
[2] Direct burial ampacities are per NEC Table 310.60(C)(83) for three insulated copper conductors rated 105°C, cabled within an overall covering and directly buried in earth at 20°C ambient earth temperature.
[3] For 8 kV ampacities, refer to NEC Tables 310.60(C)(71) and 310.60(C)(83) for cables listed 5001-35,000 volts.